Power Tool

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Power tools

What it is used for

A power tool uses electrical power or a battery to perform tasks such as drilling, cutting, grinding, sanding, or fastening. Correct tool selection, secure workholding, suitable accessories, and power isolation help improve work quality and reduce injury risks.

Main parts

- Tool housing protects the motor and internal components.
- Handle and grip support controlled operation.
- Trigger or switch starts and stops the tool.
- Power cable, battery, or power inlet supplies energy.
- Chuck, collet, or spindle holds the working accessory where applicable.
- Guard or shield helps contain debris and protect the operator.
- Speed or torque control adjusts operation where fitted.
- Auxiliary handle or depth guide supports control on applicable models.

Before you start

1. Only trained users should operate power tools appropriate to their task.
2. Read the manual and confirm the tool is suitable for the material and operation.
3. Inspect the housing, handle, cable, plug, battery, switch, guard, and accessory for damage.
4. Confirm that the accessory is compatible, correctly fitted, and rated for the tool's speed.
5. Secure the workpiece with a suitable clamp or vice and clear the work area.
6. Wear task-appropriate eye protection, hearing protection, and other required PPE.
7. Check the power supply, battery condition, and any required dust extraction before starting.

How to use

1. Disconnect the power source or remove the battery before fitting or changing accessories.
2. Install the correct accessory and tighten the chuck, collet, or fastening system according to the manual.
3. Adjust the guard, speed, torque, and depth settings for the task where applicable.
4. Secure the workpiece and position the tool so the accessory will not contact unintended objects.
5. Reconnect the power source and check that the trigger is released before starting.
6. Hold the tool firmly using the designated grip and auxiliary handle where provided.
7. Start the tool and allow it to reach the appropriate operating speed before applying it to the workpiece.
8. Apply controlled pressure and let the accessory perform the work without forcing the tool.

9. Release the trigger and wait for all moving parts to stop before setting the tool down.
10. Disconnect power or remove the battery before adjustments, cleaning, or clearing a jam.
11. Inspect the workpiece and tool, then clean and store the equipment safely.

⚠ Safety

- Never operate a power tool with damaged wiring, a defective switch, or a missing required guard.
- Keep hands, clothing, hair, and loose objects away from moving accessories.
- Do not exceed the rated speed of the tool or accessory.
- Disconnect power before changing accessories or making adjustments.
- Do not use an electric tool in wet conditions unless specifically designed and approved for them.
- Keep bystanders clear and use suitable dust extraction or respiratory protection where required.

Care & storage

- Remove dust and debris from the tool after use using the approved cleaning method.
- Inspect accessories, guards, cables, battery connections, and handles regularly.
- Keep ventilation openings clear and avoid exposing the tool to excessive moisture.
- Store batteries and tools according to manufacturer instructions.
- Arrange authorised repair if the tool overheats, sparks abnormally, vibrates excessively, or fails to stop correctly.

Common mistakes to avoid

- Using an accessory that is incompatible or rated for a lower speed than required.
- Holding the workpiece by hand instead of securing it.
- Removing guards or using the tool with damaged cables.
- Changing accessories while the power source remains connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.